

$$AA^T = AA^T$$

内积空间

$$F = \mathbb{R} / \mathbb{C}$$

内积(点积)的定义

$$\langle \cdot \rangle \quad v \times v = \begin{pmatrix} \alpha & \beta \end{pmatrix} \quad \alpha, \beta \in V$$

$$1) \quad \forall v \in V, \langle v, v \rangle \geq 0 \quad \text{正性}$$

$$2) \quad \langle v, v \rangle = 0 \Rightarrow v = 0$$

$$3) \quad \text{线性(第一个分量)}$$

$$4) \quad \text{线性(第二个分量)}$$

$$\bullet \langle v_1 + v_2, u \rangle = \langle v_1, u \rangle + \langle v_2, u \rangle$$

$$\bullet \langle \lambda v, w \rangle = \lambda \langle v, w \rangle \quad \text{共轭对称}$$

$$14) \quad \langle v, w \rangle = \overline{\langle w, v \rangle}$$

$$\text{若 } \langle v, v \rangle = a + bi, a, b \in \mathbb{R}$$

$$\overline{\langle w, v \rangle} = a - bi$$

$$F = \mathbb{R}, F = \mathbb{C} \quad \langle v, w \rangle \in \mathbb{C}$$

$$\text{例: } V = \mathbb{C}^n = \left\{ (z_1, z_2, \dots, z_n) \mid z_i \in \mathbb{C} \right\}$$

$$= \mathbb{C} \times \mathbb{C} \times \dots \times \mathbb{C}$$

$$\langle (z_1, \dots, z_n), (w_1, \dots, w_n) \rangle$$

$$= z_1 \overline{w_1} + \dots + z_n \overline{w_n}$$

$$= (z_1, z_2, \dots, z_n) \begin{pmatrix} \overline{w_1} \\ \overline{w_2} \\ \vdots \\ \overline{w_n} \end{pmatrix}$$

$$\langle (z_1, \dots, z_n), (z_1, \dots, z_n) \rangle = |z_1|^2 + \dots + |z_n|^2$$

$$z_i \overline{z_i} = |z_i|^2$$

$$f(x), g(x) \in C[a, b]$$

$$\langle f(x), g(x) \rangle = \int_a^b f(x) g(x) dx$$

$$\oint_{\mathbb{R}} f(x) g(x) dx$$

$$\langle f(x), g(x) \rangle = \int_{-\infty}^{\infty} f(x) g(x) e^{-x^2} dx$$

内积空间: 带有内积的线性空间

称之为内积空间

$$F^* \quad \langle (z_1, \dots, z_n), (w_1, \dots, w_n) \rangle = \sum_{i=1}^n z_i \overline{w_i}$$

$$C[a, b] \quad \langle f(x), g(x) \rangle = \int_a^b f(x) g(x) dx$$

$$\text{内积性质: } \langle 0, w \rangle = 0, \text{ 且 } \forall \langle \cdot, w \rangle \in V^*$$

$$\langle 0, v, u \rangle = 0 \quad \langle v, u \rangle = 0$$

$$\langle u, v \rangle = \overline{\langle v, u \rangle} = \overline{0} = 0$$

$$\langle v, u_1 + u_2 \rangle = \langle v, u_1 \rangle + \langle v, u_2 \rangle$$

$$\langle u_1 + u_2, v \rangle = \langle u_1, v \rangle + \langle u_2, v \rangle$$

$$= \langle v, u_1 \rangle + \langle v, u_2 \rangle$$

$$\langle v, \lambda u \rangle = \overline{\langle \lambda u, v \rangle}$$

$$= \overline{\lambda \langle u, v \rangle} = \overline{\lambda} \overline{\langle u, v \rangle}$$

$$= \overline{\lambda} \langle v, u \rangle$$

范数 = 长度

$$V/F, \langle \cdot, \cdot \rangle$$

$$\forall v \in V \quad \|v\| = \sqrt{\langle v, v \rangle}$$

$$\|(z_1, z_2, \dots, z_n)\| = \sqrt{|z_1|^2 + |z_2|^2 + \dots + |z_n|^2}$$

$$C[-1, 1] \quad \|f(x)\| = \sqrt{\int_{-1}^1 f(x) dx}$$

$$\text{性质: } \begin{cases} \|0\| = 0 \Rightarrow v = 0 \\ \|v\| \geq 0 \\ \|\lambda v\| = |\lambda| \|v\| \end{cases}$$

$$\| \lambda v \| = \sqrt{\langle \lambda v, \lambda v \rangle}$$

$$= \sqrt{\lambda \overline{\lambda} \langle v, v \rangle}$$

$$= |\lambda| \|v\|$$

$$= |\lambda| \|v\|$$

$$= |\lambda| \|v\|$$

$$\text{Arcos } \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} = \angle(u, v)$$

⊥ $\mathbb{R}^n$

$$\text{垂直 } u \perp v \Leftrightarrow \langle u, v \rangle = 0$$

$$v \perp u \Leftrightarrow \langle v, u \rangle = 0$$

$$0 \perp v$$

$$\text{性质: } v \perp v \Leftrightarrow v = 0$$

勾股定理 $\leftarrow$ 毕达哥拉斯定理

$$u \perp v, \|u+v\|^2 = \langle u+u, u+v \rangle$$

$$= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$$

$$\underbrace{\langle u, v \rangle + \langle v, u \rangle}_{=0}$$

$$= \|u\|^2 + \|v\|^2$$

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$$

$u, v \in V, v \neq 0$
 $u \perp v \Leftrightarrow \langle u, v \rangle = 0$
 $u \perp v \Leftrightarrow \langle u, v \rangle = 0$

证明: 假设 $u = cv + w$

$$v \perp w$$

$$\langle u, v \rangle = \langle cv + w, v \rangle$$

$$= c \langle v, v \rangle + \langle w, v \rangle$$

$$\Rightarrow c = \frac{\langle u, v \rangle}{\langle v, v \rangle} = \frac{\langle u, v \rangle}{\|v\|^2}$$

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + w$$

$$= \langle u, \frac{v}{\|v\|} \rangle \frac{v}{\|v\|} + w$$

$$\frac{v}{\|v\|} = \frac{v}{\sqrt{v \cdot v}}$$

$$u = c_1 v + u_1$$

$$= c_2 v + u_2$$

$$(c_1 - c_2)v = u_2 - u_1$$

$$\langle (c_1 - c_2)v, (c_1 - c_2)v \rangle = 0$$

$$\langle u_1 - u_2, v \rangle = 0$$

$$\Rightarrow c_1 = c_2$$

正交性

$$① v=0 \quad |\langle u, v \rangle| = 0 \neq 0 = \|u\| \|v\|$$

$$② v \neq 0 \quad |\langle u, v \rangle| = \frac{\langle u, v \rangle}{\|v\|} \cdot \|v\|$$

$$\|u\|^2 \|v\|^2 = \|u\|^2$$

$$= (\|u\|^2 \|v\|^2 + \|u\|^2)$$

$$= \frac{\langle u, v \rangle^2}{\|v\|^4} + \|u\|^2$$

$$\geq \frac{\langle u, v \rangle^2}{\|v\|^2}$$

$$\text{Cauchy-Schwarz 不等式}$$

$$\|u\| \|v\| \geq |\langle u, v \rangle|$$

证明:

$$x_i, y_i \in \mathbb{R}, \lambda_i > 0$$

$$\| \lambda_1 x_1 y_1 + \lambda_2 x_2 y_2 + \dots + \lambda_n x_n y_n \|^2$$

$$\leq (\lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2)$$

$$(\lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2)$$

$$\langle (\lambda_1 \dots \lambda_n)(y_1 \dots y_n) = \lambda_1 x_1 y_1 + \dots + \lambda_n x_n y_n$$

$$\text{由 Cauchy-Schwarz 不等式}$$

$$\|u+v\| \leq \|u\| + \|v\|$$

$$\text{等号成立} \Leftrightarrow u = cv, \quad c \geq 0$$

$$\text{证明: } \|u+v\|^2 = \langle u+v, u+v \rangle$$

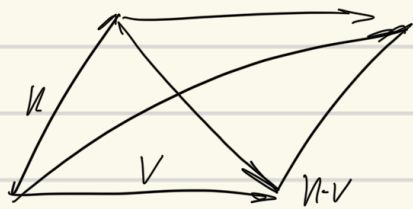
$$= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$$

$$= \|u\|^2 + \|v\|^2 + 2 \operatorname{Re} \langle u, v \rangle$$

$$\leq \|u\|^2 + \|v\|^2 + 2 |\langle u, v \rangle|$$

$$\leq \|u\|^2 + \|v\|^2 + 2 \|u\| \|v\|$$

$$= (\|u\| + \|v\|)^2$$



$$\|u+v\|^2 = \|u\|^2 + \|v\|^2 + 2\langle u, v \rangle$$

单位正交基

规范正交基

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

性质: $e_1, \dots, e_n$ 是规范正交组

$$\|a_1 e_1 + \dots + a_n e_n\|^2 = |a_1|^2 + |a_2|^2 + \dots + |a_n|^2$$

6.26 $v_1, \dots, v_m$ 是非零的, 两两垂直

$\Rightarrow v_1, \dots, v_m$ 是线性无关的

$$k_1 v_1 + \dots + k_m v_m = 0 \Rightarrow k_i = 0$$

$$0 = \langle v, v_i \rangle = k_i \langle v, v_i \rangle = k_i \|v_i\|^2$$

$$\Rightarrow k_i = 0$$

性质: $\dim V = n$

$v_1, \dots, v_m$ 规范正交组

则 $m \leq n$

定义: 若 $\dim V = n$, $e_1, \dots, e_n$

是规范正交基, 则称之为

规范正交基

$$F^n \quad e_i = (0, \dots, 0, \underset{i}{1}, 0, \dots, 0)$$

$$(e_1, e_2, e_3) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right)$$

6.3 | Gram-Schmidt

设 $v_1, \dots, v_m$ 是 V 中线性无关向量组

令 $e_1 = \frac{v_1}{\|v_1\|}$, $1 \leq n \leq m$

$$e_j = \frac{v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_{j-1} \rangle e_{j-1}}{\|v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_{j-1} \rangle e_{j-1}\|}$$

$$\text{Span} \{e_1, \dots, e_j\}$$

$$= \text{Span} \{v_1, \dots, v_j\} \quad j=1, 2, \dots, m$$

证明:

$$\text{Span} \{e_1\} = \text{Span} \left\{ \frac{v_1}{\|v_1\|} \right\} = \text{Span} \{v_1\}$$

$j=k$

$$\text{Span} \{v_1, \dots, v_k\} = \text{Span} \{e_1, \dots, e_k\} \quad \text{因为 } \langle e_i, e_j \rangle = \delta_{ij}$$

$$v_{k+1} = v_{k+1} - \langle v_{k+1}, e_1 \rangle e_1 - \dots - \langle v_{k+1}, e_k \rangle e_k$$

$$e_{k+1} = \frac{v_{k+1}}{\|v_{k+1}\|} \quad v_{k+1} \neq 0, v_{k+1} \in \text{Span} \{e_1, \dots, e_k\}^\perp$$

$$= \text{Span} \{v_1, \dots, v_{k+1}\}$$

矛盾

$$\text{Span} \{e_1, \dots, e_k, e_{k+1}\}$$

$$= \text{Span} \{v_1, \dots, v_k, v_{k+1}\}$$

$$e_{k+1} \in \text{Span} \{e_1, \dots, e_k, v_{k+1}\} = \text{Span} \{e_1, \dots, e_k\} + \text{Span} \{v_{k+1}\}$$

$$= \text{Span} \{v_1, \dots, v_k\} + \text{Span} \{v_{k+1}\}$$

$$= \text{Span} \{v_1, \dots, v_{k+1}\}$$

$$\text{Span} \{e_1, \dots, e_{k+1}\} \subseteq \text{Span} \{v_1, \dots, v_{k+1}\}$$

$$v_{k+1} = \|v_{k+1}\| e_{k+1} = \langle v_{k+1}, e_1 \rangle e_1 + \dots + \langle v_{k+1}, e_k \rangle e_k$$

$$T \in \mathcal{L}(V)$$

线性变换

$$(T v_1, T v_2, \dots, T v_n) = (v_1, v_2, \dots, v_n)$$

$$\begin{pmatrix} a_{11} & & \\ & \ddots & \\ 0 & & a_{nn} \end{pmatrix} A$$

$$(v_1, \dots, v_n) T (e_1, \dots, e_n) = \begin{pmatrix} a_{11} & & \\ & \ddots & \\ 0 & & a_{nn} \end{pmatrix} T$$

$$T(e_1, \dots, e_n) = (e_1, \dots, e_n) T^T A^T$$

上三角矩阵

$$v_1 = \|v\| e_1$$

$$v = (e_1 \dots e_n) \begin{pmatrix} \|v\| \\ 0 \\ \vdots \\ i \\ 0 \end{pmatrix}$$

$$v_2 = \|v_2'\| e_2 = \|v_2'\| \langle v_2, e_1 \rangle e_1$$

$$v_2 = \|v_2'\| e_2 + \langle v_2, e_1 \rangle e_1$$

$$v_2 = (e_1 \dots e_n) \begin{pmatrix} \langle v_2, e_1 \rangle \\ \|v_2'\| \\ 0 \\ \vdots \\ i \\ 0 \end{pmatrix} \quad v_j = \|v_j'\| e_j + \dots$$